

10 (a) (i) Temperature is a measure of the average kinetic energy of a body B1

Heat is the thermal energy that is transferred between bodies depending on the temperature difference. B1

(ii) The metal conducts heat away from the body B1  
which gives the sensation that it is cool. B1

(b) (i) 1. Check whether the mass is changing at a constant rate/bubbles produced at a constant rate. B1

2. To allow/cancel out/eliminate/compensate for heat losses (to atmosphere) B1

(ii) use of power  $\times$  time = mass  $\times$  specific latent heat C1

$$(70 - 50) \times 5 \times 60 = (13.6 - 6.5) \times L$$

$$L = 845 \text{ J g}^{-1}$$

(c) (i) An ideal gas is a hypothetical gas which obeys the equation of state of an ideal gas,  $pV = nRT$ , for all values of pressure  $p$ , temperature  $T$ , volume  $V$  and number of moles  $n$ . B1

(An ideal gas has no intermolecular forces and therefore its internal energy consists of only kinetic energy.)

(ii) It is the sum of a random distribution of kinetic energies (due to the random motion) associated with the molecules of the system. B1

(iii) 1. Heat gained by some gas = heat lost by some gas

$$\begin{aligned} n_1 c(\theta_f - 300) &= n_2 c(450 - \theta_f) \\ \frac{p_1 V_1}{RT_1} c(\theta_f - 300) &= \frac{p_2 V_2}{RT_2} c(450 - \theta_f) \end{aligned} \quad \text{C1}$$

$$\begin{aligned} \frac{(1.75 \times 10^5)(16.8 \times 10^{-3})}{(8.31)(300)} c(\theta_f - 300) \\ = \frac{(2.25 \times 10^5)(22.4 \times 10^{-3})}{(8.31)(450)} c(450 - \theta_f) \end{aligned}$$

$$\theta_f = 380 \text{ K} \quad \text{A1}$$

2.

$$n_1 + n_2 = \text{final } n$$

$$\frac{p_1 V_1}{RT_1} + \frac{p_2 V_2}{RT_2} = \frac{p_f V_f}{RT_f}$$

$$\frac{(1.75 \times 10^5)(16.8 \times 10^{-3})}{(8.31)(300)} + \frac{(2.25 \times 10^5)(22.4 \times 10^{-3})}{(8.31)(450)}$$

$$= \frac{p_f(39.2 \times 10^{-3})}{(8.31)(380)}$$

M1  
A1

$$p_f = 2.04 \times 10^5 \text{ Pa}$$

(d) (i) 0 J since cyclic process and gas goes back to same point. A1

(ii)  $W = p\Delta V$   
 $= (4.0 \times 10^5)(6.0 \times 10^{-4})$   
 $= 240 \text{ J}$ 
M1  
A1

(iii)

| change | work done on gas<br>/ J | heating supplied<br>to gas / J | increase in<br>internal energy / J |
|--------|-------------------------|--------------------------------|------------------------------------|
| P → Q  | 240                     | -600                           | -360                               |
| Q → R  | 0                       | +720                           | 720                                |
| R → P  | -840                    | +480                           | -360                               |